Lab Assignment 4

MTH 308: Numerical Analysis and Scientific Computing–I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: February 06, 2026

This lab will have you implement Gaussian elimination with pivoting to solve linear systems. Please write a single script file `Lab4LastnameRollNumber.m` and publish it as a PDF file. Then submit the script `Lab4LastnameRollNumber.m` along with the corresponding published file `Lab4LastnameRollNumber.pdf` (for example, my submitted files would be `Lab4Biswas260102.m` and `Lab4Biswas260102.pdf`). Each solution should:

- Be contained in a separate cell including the problem number and short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

1. Write a function to solve a system of equations Ax = b, using Gaussian elimination with *partial pivoting* (assuming there is a solution). Use the function template:

```
function x = Gaussian_elimination_partial_pivoting(A, b)
- code to return the solution, x, via partial pivoting -
end
```

Test your function to solve the system:

$$
\begin{aligned}
2x_1 + 0x_2 + x_3 - x_4 &= 6,\\
6x_1 + 3x_2 + 2x_3 - x_4 &= 15,\\
4x_1 + 3x_2 - 2x_3 + 3x_4 &= 3,\\
-2x_1 - 6x_2 + 2x_3 - 14x_4 &= 12.
\end{aligned}
$$

2. Similar to Q1, write a function - `Gaussian_elimination_scaled_partial_pivoting(A, b)` to solve a system using Gaussian elimination with scaled partial pivoting.

Test your function to solve the system:

$$
\begin{aligned}
\pi x_1 - e x_2 + \sqrt{2}\, x_3 - \sqrt{3}\, x_4 &= \sqrt{11},\\
\pi x_1 + e x_2 - e^2 x_3 + \tfrac{3}{7} x_4 &= 0,\\
\sqrt{5}\, x_1 - \sqrt{6}\, x_2 + x_3 - \sqrt{2}\, x_4 &= \pi,\\
\pi^3 x_1 + e^2 x_2 - \sqrt{7}\, x_3 + \tfrac{1}{9} x_4 &= \sqrt{2}.
\end{aligned}
$$

Note: For the selection of pivot elements in each method, output if a row swap was performed (including which rows were swapped) or not.